

ard_ K210 Autonomous Learning Classification

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1.K210 and Arduino communication

1.1 Experimental premises

This tutorial uses arduino, and K210 requires running the program in **K210-AI (stm32_pico_arduino)** to start the experiment

arduino *1

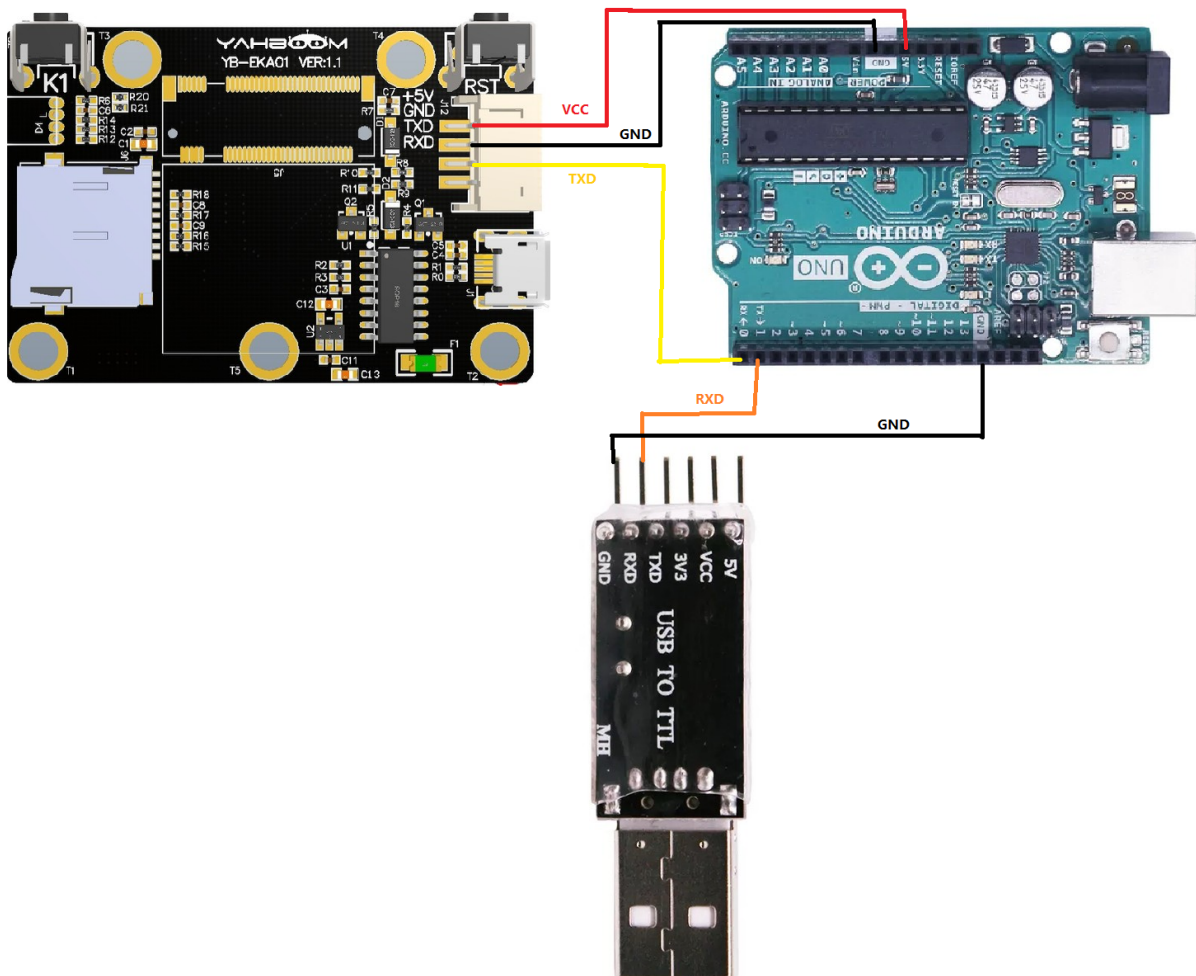
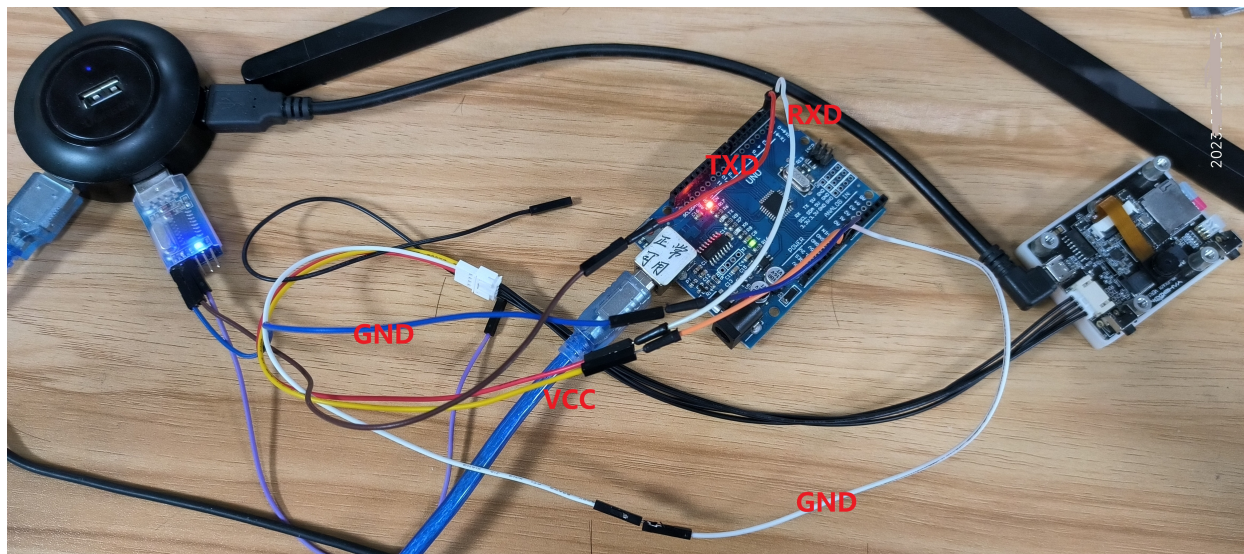
K210 perspective module * 1 (requires SD card (with AI model inside) and camera)

USB to TTL module * 1

1.2 Experimental wiring

arduino	usb to ttl
TXD	RXD
GND	GND

arduino	K210 perspective module
RXD	TXD
GND	GND
VCC	5V
Wiring as shown in the diagram:	



1.3 Main code explanation

```
void loop()
{
    while (K210Serial.available())
    {
```

```

recv_k210msg(k210Serial.read());

if (k210_msg.class_n != 0)
{
    if(k210_msg.class_n == 10)
    {
        sprintf(buff_com,"id = %c\r\n",k210_msg.id);
        k210Serial.print(buff_com);

        k210_msg.class_n = 0;
    }

}

}
}

```

After the above program, if you are running this routine, k210_ The members of the msg structure have corresponding values and are processed through serial port printing

K210_ Msg: is a structure that receives information, and its main members are

- X: is the horizontal coordinate of the top left corner of the recognized box (range: 0-240)
- Y: is the vertical coordinate of the upper left corner of the identified box (range: 0-320)
- W: is the width of the recognized box (range: 0-240)
- H: The length of the recognized box (range: 0-320)
- ID: is the recognized label
- Class_ n: Routine number
- Msg_ Msg [20]: Valid data

After receiving and processing data, k210_ Each member of the msg will store valid information.

If you want to develop it again, call K210 directly_ Members of msg are sufficient

1.4 experimental phenomena

1. After connecting the cable, the K210 perspective module runs offline. Please check 【6.2 K210 as coprocessor】 -- 【ReadMe】

2. Set the serial port assistant to the interface shown in the figure

COM Configs

Channel: COM1 #
Baudrate: 115200
Paritybits: NONE
Databits: 8
Stopbits: 1
Flowctrl: NONE
Close

Recv Options

☒ ASCII ☐ HEX
☒ Log display mode
☒ Auto linefeed
☐ Hide received data
☐ Recv save to file...
[AutoScroll](#) [Clear](#)

Send Options

☒ ASCII ☐ HEX
☒ Enable escape chars
☒ AT CMD auto CRLF
☐ Auto append bytes
☐ Send from file ...
☐ Period: 1000 ms
[Shortcut](#) [History](#)

3. Then run the object detection routine, and the serial assistant will print out the important information transmitted from k210 to stm32, as shown in the following figure

```
id = 3
id = 2
id = 1
id = 2
id = 2
id = 2
id = 1
id = 1
id = 1
id = 2
id = 2
id = 2
id = 2
id = 2
id = 2
id = 2
id = 3
id = 2
id = 1
id = 1
id = 1
id = 2
id = 1
id = 1
id = 2
```

Object detection only transmits k210_ The id of the msg is a Member variable.

ID: Refers to the object ID when taking a photo, and the range of IDs (1-3)